



FADO

Meeting 2

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Outline

1. **The λ controller:** New objective function;
2. **Ablations:** Experiments on compas with new drift generator;
3. **Weekly reviews:** KANs;
4. **The KAM-KAN?** maybe.

We change the objective function from $\mathcal{L} + \lambda \mathcal{P}(\theta)$ to $\min_{\theta} \mathcal{L}(\theta)$ s.t. $DP(\theta) \leq \epsilon$ so we can ensure more flexibility during drift and avoid the need for a per dataset tunable hyperparameter λ

Theorem 1

$\min_{\theta} \mathcal{L}(\theta)$ s.t. $DP(\theta) \leq \epsilon$ is equivalent to $\min_{\theta} \max_{\mu \geq 0} \mathcal{L}$

Proof.

Let $DP = \max_g |D_g|$, $D_g = \hat{r} - r_g$ where r_g is group g 's positive-prediction rate and $\hat{r}$ is the mean of those rates (e.g. with two groups deviations are always mirror images meaning $D_0 = -D_1$). Using the fact that $|x| \leq \epsilon \equiv -x \leq \epsilon \wedge x \leq \epsilon$ then we can rewrite $|D_g| \leq \epsilon$ as $D_g \leq \epsilon$ and $D_g \geq -\epsilon$: Same requirement without absolute value and, additionally, we know which side is being violated.

We can now rewrite the Loss function using the Lagrange multiplier μ as follows: $\mathcal{L}(\theta, \mu) = \mathcal{L}(\theta) + \sum_g (\mu_{g+}(D_g - \epsilon) + \mu_{g-}(-D_g - \epsilon))$ with $\mu_k \geq 0, \forall \mu_k$.

If we assume a fair classifier where $DP - \epsilon \leq 0$ then $\mu(DP - \epsilon) \rightarrow -\infty$ as μ increases, whereas if we assume an unfair classifier where $DP - \epsilon > 0$ then $\mu(DP - \epsilon) \rightarrow +\infty$.

Therefore, the Lagrange multiplier μ will be maximized when the classifier is unfair and minimized when the classifier is fair. This means that the optimization problem can be reformulated as a min-max problem, where we minimize the loss function with respect to the model parameters θ while maximizing it with respect to the Lagrange multiplier μ (if we instead chose to minimize the μ player would just choose 0 everytime). This leads to the equivalent formulation:

$$\min_{\theta} \max_{\mu \geq 0} \mathcal{L}(\theta, \mu).$$

□

This also means we can find the price, μ , by dual ascent making λ adaptative, dataset agnostic and not requiring hyperparameter tuning. This is a more flexible approach that allows the model to adapt to changing conditions and ensure fairness without the need for manual tuning of λ . The dual ascent finds the optimal value of μ in the following way:

$$\frac{\partial \mathcal{L}}{\partial \mu_{g\pm}} = \pm(D_g - \epsilon).$$

Updating μ as: $\mu_{g\pm} = \text{clip}(\mu_{g\pm} + \eta(\pm D_g - \epsilon), 0, \lambda_{\max})$ where η is the learning rate and $\lambda_{\max}$ is the maximum value of λ (which we can set to infinity). This way we keep non-negative prices and bound the duality requirement + accuracy trade-off.

How can we estimate D_g ?

We see one person at a time $(a_t, \hat{y}_t)$, which is not a meaningful rate for every group - we need to be right on average. The Horvitz-Thompson estimator states:

Let $Y_i, i = 1, \dots, n$ be an independent sample from n of $N \geq n$ distinct stratum with mean μ . Suppose π_i is the inclusion probability (i.e. sample belongs to the i -th stratum) then: $\hat{Y}_{HT} = \sum_{i=1}^n \frac{Y_i}{\pi_i}$ and mean estimate $\hat{\mu}_{HT} = \frac{1}{N} \sum_{i=1}^n \frac{Y_i}{\pi_i}$ is unbiased for μ .

In our case, we want to estimate r_g (equal it on average) and we can use the Horvitz-Thompson estimator to do so. We can estimate r_g as follows:

$$u_{t,g} = \frac{\hat{y}_{t=1}}{\pi_g} \text{ with probability } \pi_g \times r_g \text{ otherwise } 0.$$

On average:

$$\pi_g \times r_g \times u_{t,g} = r_g$$

Theorem 2

$\mathbb{E}[v] = D_g$: *The estimator is right on average.*

Proof.

Let the estimator $v_{t,g}$ be defined as $\frac{1}{G} \sum_h (u_{t,h} - u_{t,g})$. Because $\mathbb{E}[u_{t,g}] = \frac{1}{\pi_g} (\pi_g \times r_g) + 0(1 - \pi_g \times r_g) = r_g$, we have:

$$\mathbb{E}[v_{t,g}] = \frac{1}{G} \sum_h (\mathbb{E}[u_{t,h}] - \mathbb{E}[u_{t,g}]) = \frac{1}{G} \sum_h (r_h - r_g) = \hat{r} - r_g = D_g.$$

And thus we prove that the estimator is right on average. □

Limitations: Samples are independent given the model but not identically distributed (in online learning model's distribution changes over time)

Group-targeted drift, 5 seeds: FADO adapts and stays fair

`sim_y_swaps_group`: after a random onset (50–70% of the stream), every label of one class in group 0 (non-White) is flipped, always in the direction that *widens* the gap (label-rate gap $\approx +0.09 \rightarrow +0.60$). Group 1 is untouched. $\lambda_{\text{base}} = 1$, $\epsilon = 0.05$, $\eta = 0.01$, $\lambda_{\text{max}} = 10$.

Before / after the drift

Model	Acc pre $\uparrow$	Acc post $\uparrow$	DP pre $\downarrow$	DP post $\downarrow$	EO post $\downarrow$
ARF	0.660 ± 0.013	0.632 ± 0.055	0.105 ± 0.032	0.163 ± 0.035	0.119 ± 0.035
RFR	0.668 ± 0.016	0.728 ± 0.031	0.107 ± 0.024	0.173 ± 0.040	0.121 ± 0.037
Aranyani	0.673 ± 0.011	0.706 ± 0.015	0.036 ± 0.007	0.043 ± 0.021	0.046 ± 0.008
FADO	0.673 ± 0.011	0.729 ± 0.014	0.035 ± 0.007	0.038 ± 0.017	0.043 ± 0.007

Whole stream and controller

Model	Acc $\uparrow$	DP $\downarrow$	EO $\downarrow$	λ post	Acc. drifts	Resets
ARF	0.648 ± 0.027	0.128 ± 0.028	0.119 ± 0.025	—	—	—
RFR	0.695 ± 0.015	0.136 ± 0.031	0.126 ± 0.032	—	—	—
Aranyani	0.688 ± 0.006	0.039 ± 0.008	0.056 ± 0.010	1 (fixed)	—	—
FADO	0.697 ± 0.010	0.037 ± 0.006	0.055 ± 0.010	1.32 ± 0.10	1	1

Only 5 tuning seeds (11–55), one scenario: mean $\pm$ std, no significance test yet.

Folktables, 5 seeds: post-drift results

After the drift

Model	Acc $\uparrow$	DP $\downarrow$	EO $\downarrow$	ms/sample $\downarrow$
ARF	0.715 ± 0.007	0.066 ± 0.015	0.077 ± 0.014	–
RFR	0.781 ± 0.003	0.069 ± 0.015	0.057 ± 0.013	–
Aranyani	0.807 ± 0.005	0.055 ± 0.005	0.038 ± 0.004	63.4 ± 1.2
FADO	0.804 ± 0.004	0.041 ± 0.006	0.026 ± 0.006	67.3 ± 0.9

5 seeds (11–55), Folktables: mean $\pm$ std, no significance test yet.

KANs

KAN: Kolmogorov–Arnold Networks

Ziming Liu^{1,4*} Yixuan Wang² Sachin Vaidya¹ Fabian Ruehle^{3,4}
James Halverson^{3,4} Marin Soljačić^{1,4} Thomas Y. Hou² Max Tegmark^{1,4}

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Kolmogorov-Arnold Networks

Activation functions are placed on edges instead of nodes.

D^3 KAN

A denoising and data-complexity-adaptive KAN for long-term forecasting.

D^3 KAN: Dual Denoising and Data-Complexity-Adaptive Kolmogorov-Arnold Networks for Long-Term Time Series Forecasting

Minrui Wang^{1,2}, Jiwei Qin^{1,2*}, Di Wang^{3,4}, Dacheng Wang⁵,
Zhiqi Shen^{3,4}

Learning Fair Representations with Kolmogorov-Arnold Networks

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Learning Fair Representations with KANs

KANs applied to learning fair representations.

KKN: KAM-KAN Networks - A KAN for online learning under concept drift

